Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Sangkyun Cho	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University, Whiting School of Engineering			
Address	3400 North Charles Street Baltimore, MD 21218			

2. Technology Overview

Title of Proposed Technology	Mycoplasma Nucleic Acid Amplification Technique Assay for Cell and Gene Therapy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 21**(Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Industrial Biotechnology	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Cell and Tissue Therapeutics	Industrial Biotechnology	Other Industrial Biotechnology
Concept of Proposed Technology	Thermostable DNA polymerases, such as Taq, are essential for molecular diagnostics like PCR and qPCR. These enzymes are typically mass-produced in hosts like *E. coli*, which often leads to host-derived DNA contaminating the final purified product. This contamination can cause non-specific amplification and background signals, leading to false-positive results in high-sensitivity assays like mycoplasma NAT testing. Current industrial "workarounds" involve specialized PCR strategies (e.g., touchdown PCR or melting curve analysis) to differentiate between host DNA and target DNA. This proposal focuses on a fundamental solution: a proprietary technology to remove contaminating DNA at the source, enabling a "DNA-free" Taq polymerase that ensures higher technical performance and mitigates false-positive risks			

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Rationale for Joint R&D Project	**Technical Need:** Conventional mycoplasma testing is time-consuming, while current NAT-based assays are vulnerable to reagent DNA contamination and matrix inhibition. **Scientific Rigor:** Collaboration with a leading international research university is essential to strengthen scientific rigor and international credibility through independent performance verification. **Global Validation:** Joint efforts will focus on advanced assay design, analytics, and cross-validation using diverse sample matrices. **Commercial Translation:** The project will translate research into a manufacturable, QC-ready product package (enzyme and kit) aligned with major international compendial and ICH expectations.
Key Words	1. DNA-free Taq DNA polymerase 2. Mycoplasma NAT (qPCR/TaqMan) 3. Cell & Gene therapy product QC & in-process monitoring 4. False-positive risk mitigation 5. Analytical validation (ICH Q2(R2) / USP <1223>)

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**Develop DNA-free Taq DNA polymerase** that meets predefined residual-DNA acceptance criteria and matches the performance of leading commercial enzymes. **Establish a batch-release QC package** including tests for residual DNA, activity, purity, nuclease residue, and stability, alongside a controlled manufacturing workflow. **Validate a mycoplasma NAT kit & upgrade** utilizing the DNA-free Taq to achieve high sensitivity and robustness in bioprocess matrices. **Generate independent, internationally credible verification data** to support global commercialization and regulatory communications.			
R&D Target Specifications	**DNA-free Taq Polymerase:** No reproducible amplification in background tests; qPCR efficiency of 90–110%; linearity; and stability under frozen storage. **Mycoplasma NAT Kit & Validation:** Sensitivity (LoD) at or better than ~10 CFU/mL; high specificity with no false positives; and integrated inhibition monitoring. **Comparability:** Protocol parameters defined for major qPCR platforms, such as the Roche LC480. **Upgrade Mycoplasma NAT Kit:** By adding new features to the developed product, we aim to transition from a market follower to a market leader.			
R&D Plan	**Work packages (WP) and milestones (indicative):** WP1. Requirements & Design Inputs (Months 1–3) - Define user needs, target product profile (TPP), QC acceptance criteria, and validation strategy aligned to compendial/ICH principles. Milestone: Design inputs and risk assessment finalized. WP2. DNA-free Taq Manufacturing Process Development (Months 2–12) - Host DNA carryover mapping, nucleic-acid removal optimization, contamination control (materials, water, environment), and scale-relevant purification. Milestone: Pilot batches produced; residual DNA reduced to target level. WP3. Formulation & Stability (Months 6–18) - Optimize buffer/excipients; assess hot-start/lyophilization feasibility; conduct accelerated and real-time stability. Milestone: Final formulation selected; stability package initiated. WP4. Mycoplasma NAT Assay & Kit Engineering (Months 6–20) - Primer/probe and IPC design, matrix-robust protocol optimization, controls/standards configuration, platform transfer. Milestone: Engineering prototype kit completed. WP5. Analytical Validation & Independent Verification (Months 12–30) - LoD, specificity, precision, robustness, interference/inhibition; cross-site verification at Johns Hopkins. Milestone: Validation report and cross-verification dataset completed.			

	WP6. **Upgrade Mycoplasma NAT Kit** (Months 20–30) By adding new features (live/dead discrimination) to the developed product, we aim to transition from a market follower to a market leader. WP7. Productization Readiness (Months 24–36) - Manufacturing scale-up plan, batch-release QC SOPs, IFU documentation, commercialization package and partner PoC. Milestone: QC-ready prototype (TRL 7,8,9) and launch-ready documentation package.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	-Conduct independent performance verification of the mycoplasma NAT kit. -Provide advanced assay design and statistical data analysis support, including inhibition studies. -Evaluate the kit using diverse matrices and benchmark against reference products. -Contribute to joint technical reporting and white papers to enhance global credibility. - **Upgrade Mycoplasma NAT Kit** By adding new features (live/dead discrimination) to the developed product, we aim to transition from a market follower to a market leader.
Expected R&R of Korean SME Partner	-Lead the development of DNA-free Taq manufacturing processes and contamination control. -Engineer and manufacture the mycoplasma NAT kit reagents, controls, and packaging. -Conduct internal verification, manage IP strategies, and oversee commercialization planning. -Coordinate project governance, sample logistics, and cross-site study execution

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological:** A manufacturing-ready DNA-free Taq platform and an upgraded mycoplasma NAT product with reduced false-positive risks. **Economic:** Strengthened competitiveness of domestic bioprocess supplies, reducing reliance on imported premium enzymes. **Safety & Regulatory:** Improved safety decision-making for cell/gene therapy and biologics manufacturing through reliable, faster testing. **Global Reach:** Increased export potential through validation packages aligned with international ICH and compendial standards. **Global Reach:** Increased export potential through validation packages aligned with international ICH and compendial standards. **Comparative study:** By comparing mycoplasma contamination test kits distributed in the U.S. market with the developed test kit, its superior performance will be verified.

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6. References

(1) Ph. Eur. 2.6.7 Mycoplasmas; USP <63> Mycoplasma Tests (compendial expectations for mycoplasma testing).

(2) ICH Q2(R2) Validation of Analytical Procedures; USP <1223> Validation of Alternative Microbiological Methods (validation principles for alternative/NAT methods).